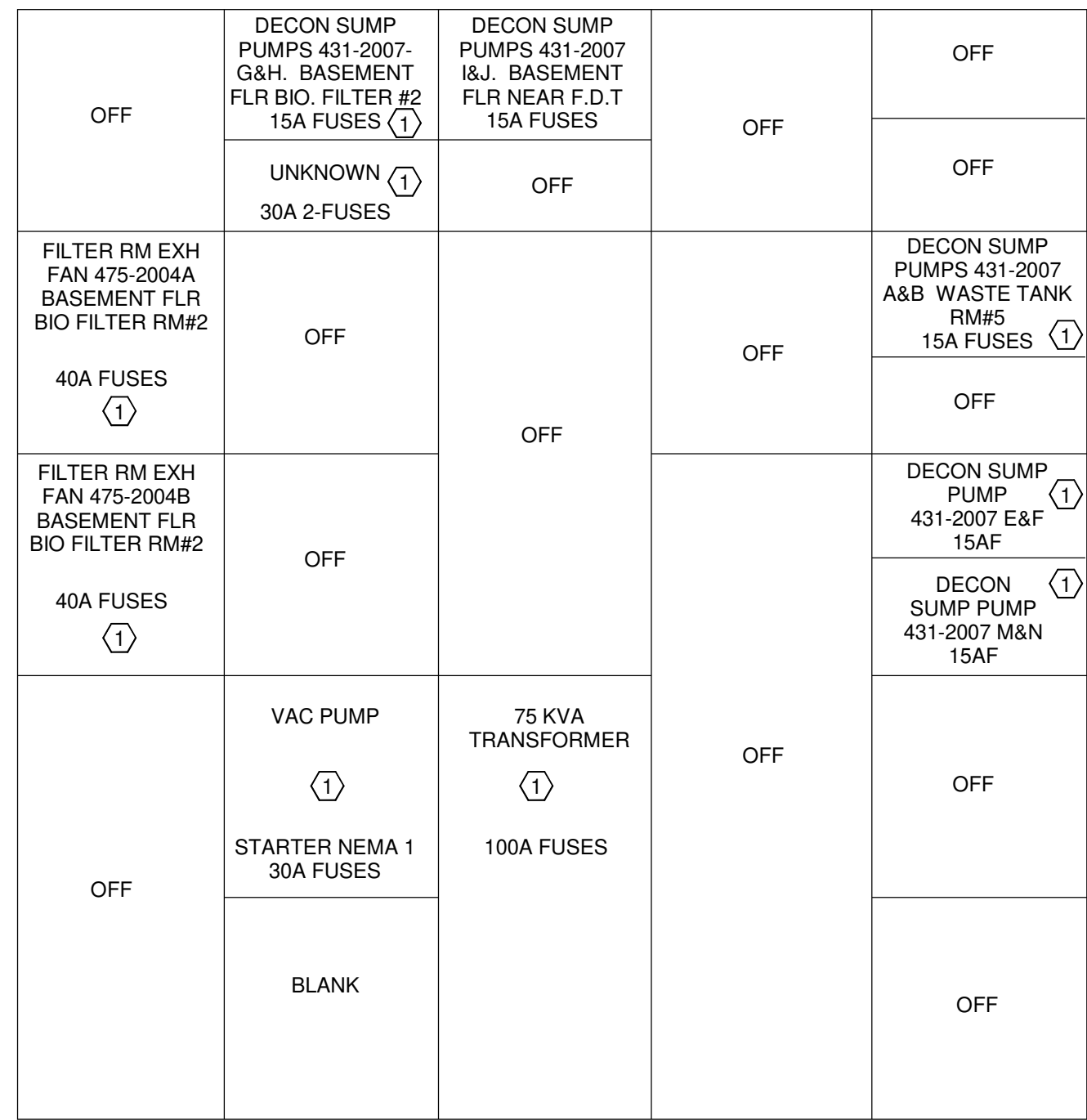




1. FURNISH SHORT CIRCUIT AND ARC FLASH STUDY WITH ARC FLASH LABELS PER SPECIFICATIONS FOR ALL NEW SWITCHBOARDS, SWITCHGEARS, PANELBOARDS, AND EXISTING SWITCHGEARS THAT THEIR EXISTING BREAKERS ARE BEING REFURBISHED.

1 TRANSFER EXISTING CIRCUIT TO THE NEW DISTRIBUTION PANEL DP62-1. FIELD VERIFY EXACT BREAKER/FUSE SIZE INDICATED ON THE ELEVATION AND MATCH. MATCH NEW WIRES AMPACITY WITH BREAKER SIZE. IF EXISTING CIRCUIT IS ROUTED THROUGH A STARTER, PROVIDE AND INSTALL COMBINATION MOTOR STARTER. MATCH EXISTING SIZE AND ROUTE CIRCUIT THROUGH. REFER TO FLOOR PLAN FOR COMBINATION MOTOR STARTER PLACEMENT LOCATION. REFER TO PANEL SCHEDULE FOR NEW BREAKERS.

② ROUTE THROUGH COMBINATION MOTOR STARTER SHOWN ON FLOOR PLAN

③ PROVIDE AND INSTALL CT'S AND CONNECT TO EXISTING CAMPUS NETWORK SYSTEM.
MATCH EXISTING SIEMENS DEM 2000 CT'S.

4 CONTRACTOR TO TRACE THE CIRCUIT AND NAME ACCORDINGLY.

5) TRANSFER EXISTING CIRCUIT TO THE NEW SWITCHGEAR SWG-11. VERIFY EXACT BREAKER/FUSE SIZE INDICATED ON THE ELEVATION AND MATCH. MATCH NEW WIRES AMPACITY WITH BREAKER SIZE. REFER TO PANEL SCHEDULE FOR NEW BREAKERS.

6 VERIFY WIRES AMPACITY AND SET TRIP SETTINGS TO MATCH THE WIRE AMPACITY.

7 SET 1600A MAIN BREAKER TRIP SETTING AT 800A.

2 BUILDING 62 - EXISTING MCC (FPE)
NOT TO SCALE

Branch Panel: L-C-62

Location: OUTSIDE
Supply From: SWG-62
Mounting: SURFACE
Enclosure: 3R

Volts: 120/240 Single
Phases: 1
Wires: 3

A.I.C. Rating: 10,000
Bus Rating: 100 A
MCB Rating: 70 A

Notes:

CKT	Circuit Description	Trip (A)	Poles	"A"	"B"	Poles	Trip (A)	Circuit Description	CKT	
1	GEN. BLOCK HEATER	20	2	1500	180		1	20	GEN. BATTERY CHARGER.	2
3	--	--	--		1500	0	1	20	SPARE	4
5	SPARE	20	1	0	0		1	20	SPARE	6
7	SPARE	20	1		0	0	1	20	SPARE	8
9	SPARE	20	1	0	0		1	20	SPARE	10
11										12
13										14
15										16
17										18
		Total Load:		1680 VA						
		Total Amps:		14 A						
Total Connected Load		Total Demand Factor		Estimated Demand		Panel Totals				
Heat		3000 VA		100.00%		3000 VA				
POWER, OTHER		180 VA		100.00%		180 VA				
								Total Conn. Load:		3180 VA
								Total Est. Demand:		3180 VA
								Total Conn. Current:		9 A
								Total Est. Demand Current:		13 A

SUGGESTED WORK PLAN & PHASING

THE FOLLOWING IS A SUGGESTED WORK PLAN GUIDE ONLY FOR USE BY THIS CONTRACTOR AS A REFERENCE TOOL. CONTRACTOR MUST DEVELOP AND SUBMIT THEIR OWN DETAILED WORK PLAN TO THE OWNER AND ENGINEER TO REVIEW AND APPROVAL. NO WORK SHALL BE COMMENCE UNTIL THE CONTRACTOR'S WORK PLAN HAS BEEN SUBMITTED AND IS APPROVED. CONTRACTOR SHALL COORDINATE ALL POWER OUTAGES VERY CLOSELY WITH THE OWNER PRIOR TO ANY OUTAGE. CONTRACTOR TO ACCOUNT FOR WORKING AFTER HOURS, BEFORE HOURS AND ON WEEKENDS PER OWNER'S DIRECTIONS.

1. SURVEY THE SITE AND THE BUILDING FOR EXACT LOCATIONS OF EXISTING AND NEW ELECTRICAL EQUIPMENT THAT ARE PART OF THIS PROJECT AND IDENTIFY ALL NEW AND EXISTING CONDUIT ROUTINGS.
2. INSTALL NEW SWITCHGEAR AND NEW DISTRIBUTION PANELS AS SHOWN ON THE DRAWINGS.
3. REMOVE MCC-4 OUT OF THE ROOM AS SHOWN ON THE DRAWINGS.
4. INSTALL NEW FEEDERS FROM THE NEW SWITCHGEAR OUTSIDE TO NEW AND EXISTING PANELS AS SHOWN ON THE DRAWINGS.
 - A. MAKE FINAL CONNECTION TO THE NEW PANELS THAT ARE REPLACING THE EXISTING MCC'S.
 - B. SECURE THE FEEDERS CLOSE TO THE EXISTING PANELS THAT ARE TO REMAIN, DO NOT MAKE FINAL CONNECTION AT THIS POINT.
5. PROVIDE AND INSTALL NEW FEEDERS FROM THE EXISTING TRANSFORMER TO THE NEW SWITCHGEAR AND ENERGIZE THE SWITCHGEAR:
 - A. DE-ENERGIZE THE TRANSFORMER AS NEEDED FROM THE PAD MOUNT SWITCH THAT IT IS FED FROM. BEFORE DO SO, COORDINATE WITH OWNER, START THE EXISTING GENERATOR ON SITE AND TRANSFER POWER TO THE BUILDING TO GENERATOR POWER.
 - B. PROVIDE AND INSTALL EXTRA LUGS FOR THE TRANSFORMER AS NEEDED IN ORDER TO TEMPORARY KEEP BOTH NEW AND EXISTING SWITCHGEAR ENERGIZED.
 - C. AFTER THE NEW SWITCHGEAR IS ENERGIZED, STOP THE GENERATOR AND TRANSFER POWER TO THE BUILDING TO NORMAL POWER.
6. PROVIDE TEMPORARY GENERATOR AND ALL PARTS AND CONNECTIONS NEEDED FOR CONNECTING POWER TO NEW SWITCHGEAR IN EVENT OF LOSING NORMAL POWER TO THE BUILDING WHILE THE EXISTING SWITCHGEAR AND GENERATOR ARE BEING REMOVED.
7. DISCONNECT POWER AND FEEDER TO PANELS THAT ARE SHOWN TO REMAIN ON DRAWINGS AND CONNECT NEW FEEDERS. DISCONNECT AND RE-CONNECT POWER ONE PANEL AT A TIME. COORDINATE VERY CLOSELY WITH OWNER FOR ALL POWER INTERRUPTIONS.
8. TRANSFER OVER CIRCUITS FROM EXISTING MCC'S TO NEW PANELS ONE CIRCUIT AT A TIME AS SHOWN ON DRAWINGS. AFTER ALL CIRCUITS HAVE BEEN TRANSFERRED REMOVE ALL MCC'S AND UNUSED CONDUITS AND WIRES.
9. AFTER ALL CIRCUITS FED OUT OF THE EXISTING SWITCHGEAR HAVE BEEN TRANSFERRED AND RE-FED FROM THE NEW SWITCHGEAR, REMOVE EXISTING SWITCHGEAR AND EXISTING GENERATOR.
10. INSTALL NEW GENERATOR AND FOLLOW ALL THE TESTING REQUIREMENTS.
11. REMOVE TEMPORARY GENERATOR AFTER THE NEW GENERATOR IS INSTALLED AND FUNCTIONAL.

1 ALLIED DRIVE, BUILDING 2
SUITE 2600
LITTLE ROCK, AR 72202
Phone: (501) 666-6776
Fax: (501) 663-8888
tme@bernhardtme.com



100% CONSTRUCTION
DOCUMENTS

MCC BREAKER MAINTENANCE

JEFFERSON LABORATORIES
33900 NCTR ROAD, JEFFERSON, ARKANSAS 72079

[illegible]

MARK	DATE	DESCRIPTION
ISSUE DATE:		10-15-21
PROJECT NUMBER:		01-19-0062

SHEET TITLE:

BUILDING 62 - ELEVATIONS,
SCHEDULES AND NOTES

SHEET NUMBER:

E1240



CORPORATE SEAL



ENGINEER SEAL

100% CONSTRUCTION
DOCUMENTS

MCC BREAKER MAINTENANCE

JEFFERSON LABORATORIES
3900 NCTR ROAD, JEFFERSON, ARKANSAS 72079

OFF		PNL D1 XFMR ②③ 30A3P BRK	WELD REC ②③ 30A3P BRK	OFF	TAPPED BUS FOR 225KVA XFMR FOR PANEL AA ②③	COMPRES. ②③ 30A3P BRK	XFMR AT METALS LAB 1ST FLR ②③ 30A3P BRK	OFF	CONT. LAB. DIST. PNL CL ②③ 30A3P BKR	ELEVATOR ②③ 60A3P BKR	OFF
OFF	OFF	TRANSFORMER -217- PNL IN 62A 1ST FLOOR ②③ 30A3P BREAKER		OFF	TRANSFORMER FOR PANEL A.P. ②③ 150A3P BREAKER	OFF	OFF		OFF		OFF
OFF		OFF		OFF		OFF	OFF		OFF		OFF
OFF		OFF		OFF	OFF	OFF	OFF		OFF		OFF
BLANK		OFF		OFF	OFF	OFF	OFF		OFF		OFF

1 BUILDING 62 - EXISTING MCC-7 (FPE)
NOT TO SCALE

MAIN BREAKER 400A	BSL AHU AND EMC CONTROL POWER ① 60A3P BREAKER	CONTROL TRANSFORMER	CONTROL TRANSFORMER BREAKER
	PNL CL2A ① 40A3P BREAKER	OFF	OFF
OFF	BLANK	OFF	OFF
	BLANK	EF-1 NORTH 4,5,6 ① 80A3P BREAKER	BLANK
BLANK	OFF	EF-1.1 SOUTH 1,2,3 ① 80A3P BREAKER	

2 BUILDING 62 - EXISTING MCC-A (CH)
NOT TO SCALE

Branch Panel: HDP-7															
Location: 2ND FLR ELEC. RM						Volts: 480/277 Wye				A.I.C. Rating: 35,000					
Supply From: SWG-62						Phases: 3				Bus Rating: 600 A					
Mounting: SURFACE						Wires: 4				MCB Rating: 600 A					
Enclosure: NEMA 1															
Notes: EQUAL TO SQUARE-D HCM I-LINE WITH 99" BREAKER SPACE.															
C KT	Circuit Description			Trip (A)	Poles	"A"		"B"		"C"		Poles	Trip (A)	Circuit Description	C KT
1	PANEL D1 TRANSFORMER			30	3	0	0					3	30	WELDING RECEPTACLE	2
3	--			--	--			0	0			--	--	--	4
5	--			--	--					0	0	--	--	--	6
7	TRANSFORMER-217 FOR PNL 662A 1ST FLOOR			30	3	0	0					3	400	PANEL AA 225KVA TRANSFORMER	8
9	--			--	--			0	0			--	--	--	10
11	--			--	--					0	0	--	--	--	12
13	PANEL AP TRANSFORMER			150	3	0	0					3	30	COMPRESSOR	14
15	--			--	--			0	0			--	--	--	16
17	--			--	--					0	0	--	--	--	18
19	TRANSFORMER AT METALS LAB 1ST FLR			30	3	0	0					3	30	CONT. LAB. DIST. PNL CL	20
21	--			--	--			0	0			--	--	--	22
23	--			--	--					0	0	--	--	--	24
25	ELEVATOR			60	3	0	0					3	250	* SPARE	26
27	--			--	--			0	0			--	--	--	28
29	--			--	--					0	0	--	--	--	30
31															32
33															34
35															36
37															38
39															40
41															42
Total Load:				0 VA		0 VA		0 VA							
Total Amps:				0 A		0 A		0 A							
Load Classification				Connected Load		Demand Factor		Estimated Demand		Panel Totals					
										Total Conn. Load: 0 VA					
										Total Est. Demand: 0 VA					
										Total Conn. Current: 0 A					
										Total Est. Demand Current: 0 A					

Branch Panel: HDP-A															
Location: 2ND FLR ELEC. RM						Volts: 480/277 Wye				A.I.C. Rating: 35,000					
Supply From: SWG-62						Phases: 3				Bus Rating: 600 A					
Mounting: SURFACE						Wires: 4				MCB Rating: 600 A					
Enclosure: NEMA 1															
Notes: EQUAL TO SQUARE-D HCM I-LINE WITH 99" BREAKER SPACE. *ADJUSTABLE THERMAL MAG BREAKER.															
C KT	Circuit Description			Trip (A)	Poles	"A"		"B"		"C"		Poles	Trip (A)	Circuit Description	C KT
1	BSLAHU AND EMC CONTROL POWER			30	3	0	0					3	40	PNL CL2A TRANSFORMER	2
3	--			--	--			0	0			--	--	--	4
5	--			--	--					0	0	--	--	--	6
7	EF-1 NORTH 4,5,6			80	3	0	0					3	80	EF-1 SOUTH 1,2,3	8
9	--			--	--			0	0			--	--	--	10
11	--			--	--					0	0	--	--	--	12
13	* SPARE			250	3	0						--	--	--	14
15	--			--	--			0				--	--	--	16
17	--			--	--					0		--	--	--	18
19															20
21															22
23															24
25															26
27															28
29															30
31															32
33															34
35															36
37															38
39															40
41															42
Total Load:				0 VA		0 VA		0 VA							
Total Amps:				0 A		0 A		0 A							
Load Classification				Connected Load		Demand Factor		Estimated Demand		Panel Totals					
										Total Conn. Load: 0 VA					
										Total Est. Demand: 0 VA					
										Total Conn. Current: 0 A					
										Total Est. Demand Current: 0 A					

MARK DATE DESCRIPTION

ISSUE DATE: 10-15-21

PROJECT
NUMBER: 01-19-0062

SHEET TITLE:
BUILDING 62 - ELEVATIONS,
SCHEDULES AND NOTES

SHEET NUMBER:
E1250